

Question Paper

Exam Date & Time: 27-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

THIRD SEMESTER B.TECH. DEGREE EXAMINATIONS - DEC 2025/JAN 2026
SUBJECT: ECE 2125/ECE_2125 ELECTROMAGNETIC WAVES
(SPL: ELECTRONICS AND COMMUNICATION ENGINEERING)

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) A vector field is given in Cartesian coordinates as $F = 10\mathbf{a}_x + y\mathbf{a}_y + xz\mathbf{a}_z$. Express the vector at point P ($x=2, y=3, z=4$) in Cylindrical and Spherical Polar coordinates. (4)

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1B) Find the acute angle between the two vectors $A = 2\mathbf{a}_x + \mathbf{a}_y + 3\mathbf{a}_z$ and $B = \mathbf{a}_x - 3\mathbf{a}_y + 2\mathbf{a}_z$ by using the definition of (i) the dot product; (ii) the cross product. (3)

Course Outcome: "CO1" / Cognitive Level: "Understand"

- 1C) A $2\text{-}\mu\text{C}$ point charge is located at A (4, 3, 5) in free space. Find E_ρ , E_ϕ , and E_z at P (8, 12, 2). (3)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2A) State and establish the divergence theorem in the context of electrostatics, relating to the integral and differential forms of Gauss's law. (4)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 2B) Given the potential field $V = 2x^2y - 5z$, and a point $P(-4, 3, 6)$, find V , E , the direction of E , D , and ρ_v . (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2C) Find the energy stored in free space for the region $2\text{ mm} < r < 3\text{ mm}$, $0 < \theta < 90^\circ$, $0 < \phi < 90^\circ$, given the potential field $V = (200/r)$ Volt. (3)

Course Outcome: "CO2" / Cognitive Level: "Analysis"

- 3A) Given the field $\mathbf{H} = 6r \sin\phi \mathbf{a}_r + 18r \sin\theta \cos\phi \mathbf{a}_\phi$, evaluate both sides of Stoke's theorem for the portion of the cone $\theta = 0.1\pi$ bounded by $r = 2$, $r = 4$, $\phi = 0$ and $\phi = 0.3\pi$. Let the positive direction of $d\mathbf{s}$ be $\mathbf{a}_\theta$. (4)

Course Outcome: "CO3" / Cognitive Level: "Application"

- 3B) Derive an expression for magnetic field intensity at a point due to a finite conductor carrying a (3)

current along z axis.

Course Outcome: "CO3" / Cognitive Level: "Understand"

- 3C) Let $\mu = \mu_1 = 4 \mu\text{H/m}$ in region 1 where $z > 0$, whereas $\mu_2 = 7 \mu\text{H/m}$ in region 2 where $z < 0$. (3)
Let the surface current density $\mathbf{K} = 80 \mathbf{a}_x \text{ A/m}$ on the surface $z = 0$. If magnetic flux density $\mathbf{B}_1 = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$ in region 1 find the value of $\mathbf{B}_2$ in region 2.

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4A) A uniform plane electromagnetic wave operates at a frequency $f = 600 \text{ MHz}$ and is propagating in a (4)
lossless dielectric medium with:
Relative permittivity $\epsilon_r = 4$, Relative permeability $\mu_r = 9$.
Given:
Permittivity of free space: $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m}$
Permeability of free space: $\mu_0 = 4\pi \times 10^{-7} \text{ H/m}$
Assume maximum electric field amplitude $E_0 = 200 \text{ V/m}$, polarized along x-axis, and propagation in +z-direction.
Find:
i) The propagation constant β .
ii) The wavelength λ .
iii) The equations of $E(z, t)$ and $H(z, t)$ fields.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4B) An H field in free space is given as $H(x, t) = 10 \cos(10^8 t - \beta x) \mathbf{a}_y \text{ A/m}$. (3)
Find: (a) β ; (b) λ ; (c) $E(x, t)$ at P (0.1, 0.2, 0.3) at $t = 1 \text{ ns}$.

Course Outcome: "CO4" / Cognitive Level: "Application"

- 4C) With a suitable figure and the necessary equation, explain what is Skin Depth. Explain how the (3)
depth of penetration is related to the signal attenuation.

Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5A) A 50 Ohm lossless line connects a signal of 110 KHz to a load of 75 Ohms. The load power is 100 (4)
mW. Calculate the
(i) Voltage reflection coefficient
(ii) VSWR
(iii) Position of first $V_{\min}$ and $V_{\max}$.
(iv) Impedance at $V_{\min}$ and $V_{\max}$ and the values of $V_{\min}$ and $V_{\max}$.

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5B) A transmission line has the following parameters: (3)
 $R = 5 \text{ ohm/m}$, $G = 0.5 \text{ mmho/m}$, $f = 1.5 \text{ GHz}$, $L = 8 \text{ nH/m}$ and $C = 0.23 \text{ pF/m}$.
Calculate: (a) The characteristic impedance (b) Propagation constant.

Course Outcome: "CO5" / Cognitive Level: "Application"

- 5C) Compare and contrast distributed and lumped parameter Transmission Line models, specifying the (3)
conditions under which each is applied and describing the relevant per-unit-length line parameters.

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	5	15(30%)
3	Application	7	26(52%)
4	Analysis	3	9(18%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)